Adaptive Sensor Fusion & Kalman Filtering for Fault-Tolerant Estimation

Introduction

Reliable sensor data is critical in systems like robotics, autonomous vehicles, and drones, where decisions are made in real time based on incoming measurements. However, in practical environments, sensor readings are rarely perfect. They are often affected by noise, sudden spikes, signal loss, or gradual drift over time. These issues can lead to inaccurate state estimation, unstable control, and poor system performance.

This project was developed to address these challenges by improving the reliability and accuracy of sensor measurements using sensor fusion and Kalman filtering. The idea is to move beyond relying on a single noisy sensor and instead combine multiple inputs while intelligently filtering out uncertainty. While basic sensor fusion can reduce some noise, it still struggles when the data contains faults like spikes or long-term drift. This is where the Kalman filter becomes useful, as it provides a systematic way to estimate the true state by balancing prediction and measurement under uncertainty.

To evaluate this, different real-world fault conditions were simulated, including normal noise, sudden spikes, gradual drift, and a combination of all three. These scenarios were chosen to reflect common problems faced in embedded and autonomous systems. The performance of a standard fusion approach was then compared with a Kalman filter using RMSE as a metric.

In addition, a simple drone navigation simulation using noisy GPS data was included to demonstrate how these techniques perform in a practical application. The goal was not just to reduce numerical error, but to show how improved estimation directly translates to smoother and more reliable system behavior.

Overall, this project was done to understand how robust estimation techniques can improve system performance in uncertain environments and to explore their practical relevance in real-world applications like autonomous navigation and fault-tolerant embedded systems.

What was done

- Simulated sensor data with different fault conditions:
  - Normal noise
  - Sudden spikes
  - Gradual drift
  - Mixed faults
- Applied weighted sensor fusion
- Implemented a Kalman filter for state estimation
- Compared performance using RMSE

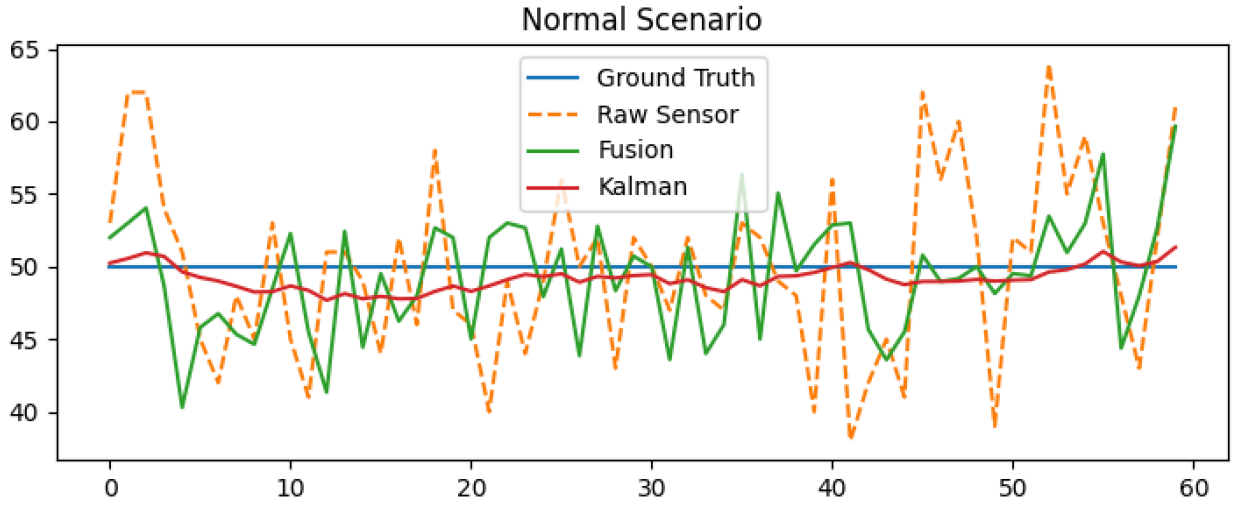
Results

Scenario	Fault Rate	Fusion RMSE	Kalman RMSE	Improvement
Normal	6.67%	4.05	1.16	71.30%
Spike	13.89%	3.94	0.77	80.48%
Drift	1.11%	5.02	2.98	40.64%
Mixed	39.44%	6.26	1.51	75.93%

Graphs

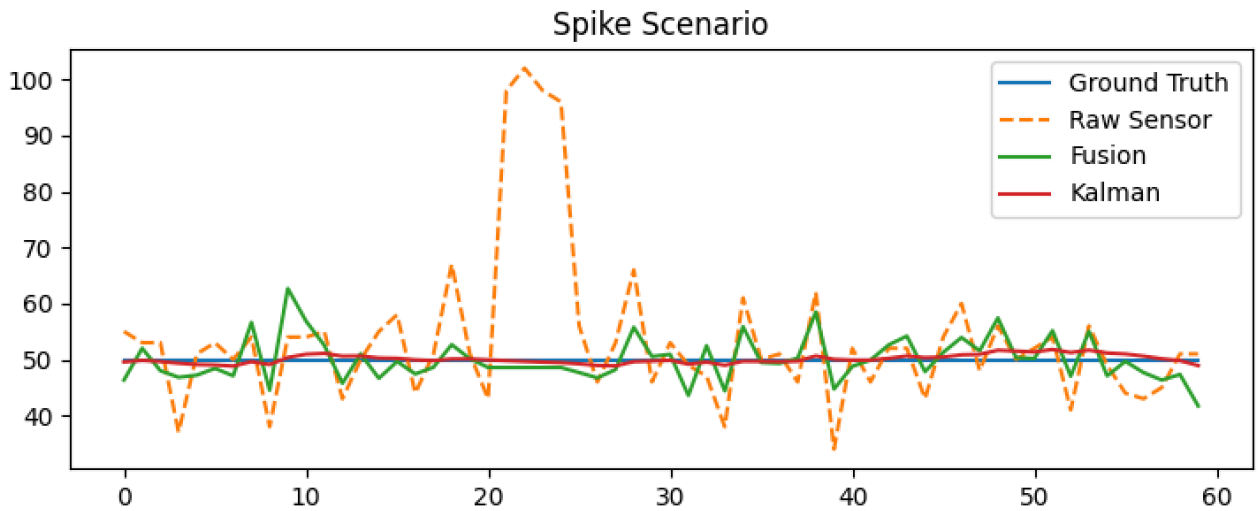
1. Normal Scenario

- Kalman output is smoother and follows the ground truth closely
- Fusion reduces noise but still fluctuates



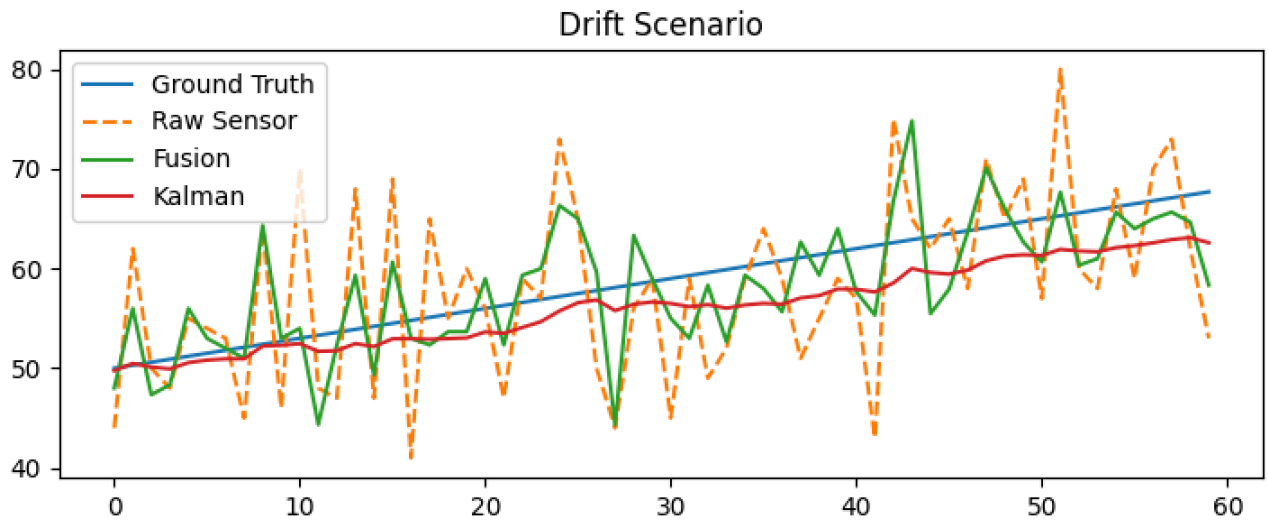
Spike Scenario

Raw sensor data contains large spikes, and fusion is still affected by these outliers.
Kalman filter suppresses sudden spikes effectively, maintaining a stable estimate.



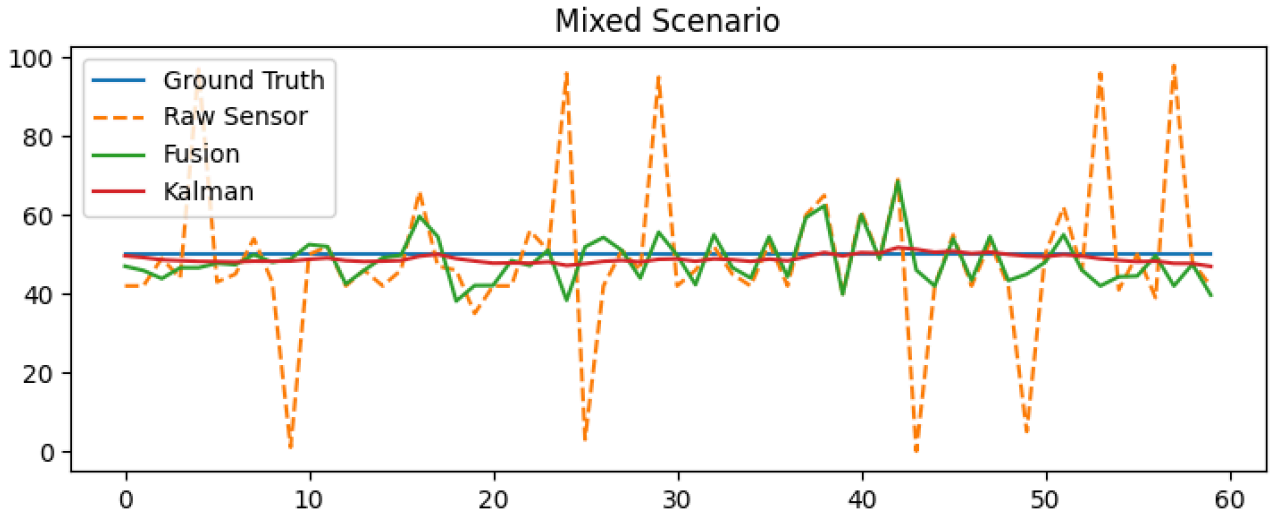
Drift Scenario

Sensor values gradually drift away from the true value, affecting overall accuracy.
Kalman filter tracks the trend better, though with slight lag in adapting to drift.



Mixed Scenario

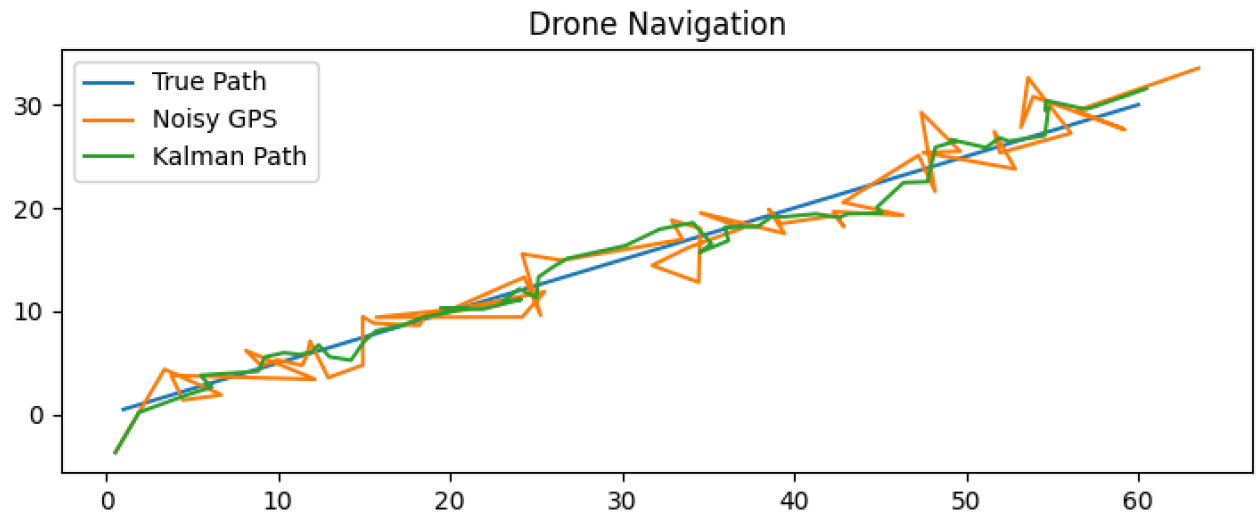
- Combination of noise, spikes, and drift causes large variations in raw and fused data.
- Kalman filter remains stable and provides a more reliable estimate under mixed faults.



Drone Navigation

- Noisy GPS data results in a scattered and inconsistent path.
- Kalman-filtered trajectory is smoother and stays closer to the true path.

Metric	Value
RMSE (Noisy GPS)	2.88
RMSE (Kalman)	1.71
Improvement	40.58%



- Noisy GPS path has visible deviations
- Kalman-filtered path is smoother
- Estimated trajectory stays closer to the actual path

Key Observations

- The Kalman filter consistently reduces estimation error across all tested scenarios compared to basic sensor fusion.
- It performs particularly well in handling sudden disturbances, showing strong robustness in spike and mixed fault conditions.
- Performance is relatively lower in drift scenarios, as gradual bias is harder to correct without additional tuning or modeling.
- In the drone simulation, improved estimation directly results in a smoother and more accurate navigation path.

Tech Stack

- Python
- NumPy
- Matplotlib

Possible Improvements

- Extend the model to an Extended Kalman Filter (EKF) for handling nonlinear systems.
- Implement adaptive tuning of process and measurement noise for better performance under varying conditions.
- Validate the approach using real-world sensor data instead of simulated inputs.
- Expand the system to support 2D/3D state estimation for more realistic applications.

Conclusion

This project demonstrates how combining sensor fusion with Kalman filtering can significantly improve the reliability of sensor-based systems operating under uncertainty. While basic fusion methods help reduce noise, they are not sufficient when dealing with real-world issues like spikes and drift. The Kalman filter provides a more structured and effective approach to estimation, resulting in lower error and more stable outputs.

Through both simulated fault scenarios and a drone navigation example, it is clear that better estimation leads to better system behavior. This makes such techniques highly relevant for applications in robotics, autonomous systems, and embedded solutions where accuracy and reliability are critical.